

WELLINGTON, New Zealand

WMO: 934360

Lat: 41.3311S

Lon: 174.8060E

Elev: 23

StdP: 14.68

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 00488

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 35.7	(c) 39.1	(d) 31.6	(e) 25.9	(f) 47.0	(g) 33.4	(h) 28.0	(i) 47.0	(j) 42.3	(k) 47.8	(l) 38.8	(m) 48.0	(n) 13.1	(o) 10	(p) 0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.8	75.1	64.4	73.0	63.4	70.8	62.7	66.9	71.8	65.5	70.3	64.2	69.1	17.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	91.6	69.0	63.1	86.6	68.0	62.4	84.2	67.5	31.4	71.7	30.4	70.2	29.5	69.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 36.1	(b) 32.0	(c) 29.2	(e) 35.0	(f) 79.6	(g) 1.9	(h) 2.4	(i) 33.7	(j) 81.3	(k) 32.5	(l) 82.7	(m) 31.5	(n) 84.0	(o) 30.1	(p) 85.7		
			DB	WB												
			33.6	69.4												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	63.2	63.4	61.3	57.9	55.1	51.3	49.7	50.5	52.9	55.1	57.7	61.6
	DBStd	6.38	5.20	5.78	5.15	4.37	4.12	3.92	3.64	3.48	3.70	3.62	3.73	3.91
	HDD50	170	5	8	6	3	7	32	52	36	15	5	1	0
	HDD65	3167	89	78	130	215	308	412	475	448	364	308	221	119
	CDD50	2574	413	384	357	240	164	69	42	53	101	162	230	359
	CDD65	974	32	34	16	1	0	0	0	0	0	0	1	13
	CDH74	102	44	35	10	0	0	0	0	0	0	0	1	12
	CDH80	2	1	0	0	0	0	0	0	0	0	0	0	1
Wind	WSAvg	15.4	16.2	15.0	14.9	14.4	14.7	14.8	14.6	14.5	15.9	16.7	16.9	16.2
Precipitation	PrecAvg	47.6	2.9	2.6	3.2	3.7	4.6	5.3	5.3	5.0	4.0	4.3	3.3	3.3
	PrecMax	63.3	5.4	13.7	6.2	7.2	7.8	8.5	10.4	8.0	7.6	10.1	7.7	7.5
	PrecMin	38.3	0.4	0.2	0.9	1.1	0.6	1.5	0.6	1.1	0.8	1.3	0.6	1.3
	PrecStd	6.3	1.4	2.6	1.5	1.6	1.9	1.7	2.3	1.8	1.6	2.2	1.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.4	77.3	75.3	69.7	66.3	61.9	60.2	60.7	63.0	66.5	70.2	75.4
		MCWB	67.1	65.2	64.4	62.6	60.2	56.8	54.9	53.9	55.3	57.7	60.5	64.0
	2%	DB	75.0	75.0	71.9	67.6	64.4	60.5	58.7	58.9	61.1	64.2	67.6	71.8
		MCWB	64.3	64.7	62.6	61.0	59.3	56.1	53.6	53.3	55.0	56.2	58.9	62.1
	5%	DB	72.0	73.0	69.9	66.0	62.7	59.0	57.1	57.4	59.4	62.5	64.8	69.7
		MCWB	63.1	63.7	62.4	60.1	57.9	54.6	52.4	52.2	54.3	56.0	57.5	61.5
	10%	DB	69.9	70.2	68.1	64.4	61.1	57.5	55.6	55.8	58.8	60.8	63.4	67.8
		MCWB	62.4	63.1	61.6	59.2	56.5	53.1	51.1	51.2	53.9	55.3	56.8	60.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.8	69.0	66.8	64.8	62.3	58.3	56.9	56.1	58.3	60.1	63.0	66.5
		MCDB	74.6	73.2	71.6	67.6	64.6	60.8	59.1	58.6	61.4	63.6	67.5	71.5
	2%	WB	66.8	67.2	65.0	62.6	60.1	56.5	54.7	54.6	56.8	58.6	60.8	64.6
		MCDB	72.0	71.6	69.9	65.8	63.0	59.5	57.5	57.6	59.9	62.2	65.1	69.1
	5%	WB	65.2	65.6	63.6	61.1	58.8	55.2	52.9	53.0	55.4	57.2	59.4	63.3
		MCDB	70.3	70.4	68.4	64.7	62.0	58.4	56.2	56.5	59.0	61.0	63.4	67.7
	10%	WB	63.7	64.2	62.4	59.6	57.2	54.0	51.8	52.0	54.3	55.8	58.2	62.0
		MCDB	68.6	69.5	67.2	63.5	60.8	57.3	55.2	55.6	58.1	59.9	62.3	66.3
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.8	10.5	9.6	8.8	8.7	8.8	9.0	9.1	9.7	10.2	10.1
		MCDBR	13.6	13.6	12.8	10.5	9.4	8.8	9.2	10.1	9.8	11.0	12.2	12.6
		MCWBR	7.3	6.9	6.9	6.9	7.0	7.2	7.1	7.2	7.0	7.3	7.6	7.1
	5% WB	MCDBR	11.9	10.8	10.3	8.6	7.9	7.5	7.7	8.3	8.7	9.3	10.0	10.2
		MCWBR	7.4	6.9	7.0	6.9	6.9	6.9	7.0	6.8	7.2	7.2	7.2	6.8
Clear-Sky Solar Irradiance	taub		0.348	0.335	0.324	0.312	0.296	0.295	0.294	0.305	0.327	0.343	0.334	0.344
	taud		2.547	2.606	2.628	2.650	2.662	2.634	2.635	2.616	2.536	2.506	2.556	2.563
	Ebn at Noon		311	307	297	280	264	252	261	277	290	301	313	314
	Edh at Noon		34	30	27	23	19	18	19	23	29	33	33	34
All-Sky Solar Radiation	RadAvg		2115	1864	1460	969	656	489	565	825	1204	1622	2008	2088
	RadStd		135	159	87	69	38	29	40	47	75	110	156	143

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.95	N/A	N/A	N/A	N/A	-41	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page